IGARSS 2021 - 2021 IEEE International Geoscience and Remote Sensing Symposium | 978-1-6654-0369-6/21/$31.00 ©2021 IEEE | DOI: 10.1109/IGARSS47720.2021.9553342

SAR AUTOMATIC TARGET RECOGNITION BASED ON MULTI-SCALE CONVOLUTIONAL FACTOR ANALYSIS MODEL WITH MAX-MARGIN CONSTRAINT

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ABSTRACT

In this study, a multi-scale convolutional factor analysis model with max-margin constraint (MMCFA) is developed for synthetic aperture radar (SAR) automatic target recognition. Compared with the traditional factor analysis (FA) model, the CFA model can maintain the spatial correlation among the image pixels in two-dimensional space and capture the structural information from images via convolution kernels. Moreover, multi-scale convolution kernels are adopted to capture richer features at different scales of SAR. Since the unsupervised model may not offer discriminative factors for the SAR target recognition task, it is expected to introduce the supervised information to the multi-scale CFA model when supervised information is available. Thus, a latent variable support vector machine (LVSVM) is linked to the factors learned from multi-scale CFA model, yielding max-margin discrimination, learned with the multi-scale CFA model jointly in a united framework. Experimental results on MSTAR dataset show that the proposed model has excellent recognition performance.

Index Terms— Factor analysis (FA), max-margin learning, multi-scale convolution kernels, synthetic aperture radar (SAR) automatic target recognition

1. INTRODUCTION

Nowadays, SAR images are becoming a useful tool in target recognition [1], [2], and automatic target recognition (ATR) of SAR images is of significant value both in commercial and military areas. A great many of methods, which can be generally divided into two kinds, have been applied to ATR of SAR images. One is devoted to represent the objects in another subspace in order to obtain discriminative features. For example, Principal Component Analysis (PCA) [3], Sparse Representation (SR) [2], [4], Graphic models [5] and scattering center models [6] are such kind of methods. The other one aims at using different classification rules to achieve better results such as Template Matching [7], Particle Swarm Optimization (PSO) with Hausdorff Distance (HD) [8], Adboost classifier [9] etc. These methods have gained good accuracies on SAR image recognition. However, all these traditional algorithms neglect the human learning system, which is able to learn representations of the objects with discriminative information.

Statistical generative models have received intensive attention from data analysis community [10]. By projecting every observation into a low dimensional latent space, statistical generative models could capture some abstract feature from the data, e.g. the factors in the factor analysis (FA) [10] model. In many previous work [10], the FA model has performed very well in high-resolution range profile (HRRP) target recognition. Nevertheless, when using the FA model to SAR target recognition, since the two-dimensional data should be stretched into vectors in FA models [10], only one-dimensional correlation of SAR images pixels can be considered while the two-dimensional correlation is ignored. In image processing researches, to maintain the spatial correlation among the image pixels, there has been significant recent interest in convolutional models [11]. One of the key characteristics of these algorithms is the exploitation of convolution operator, which plays an important role in analyzing the correlation among image pixels in two-dimensional space. Moreover, the size of convolution kernels is much smaller than the size of images, leading to the fact that features extracted by these models focus more on the local details.

In this paper, we develop a multi-scale convolutional factor analysis model with max-margin constraint (MMCFA) for SAR automatic target recognition. We refer to the FA model exploiting the two-dimensional convolution operation as the convolutional factor analysis (CFA) model. The learned convolution kernels can notice the correlation between the SAR image pixels in local regions, thus solving the problem that the traditional FA model ignores the spatial correlation among the SAR image pixels. In addition, multi-scale convolution kernels are also employed in our CFA model to extract features at different levels jointly. The kernels at smaller scales mainly capture lower-level features,

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This work was supported in part by the National Science Foundation of China under Grant 61771362, in part by the 111 Project under Grant B18039. (Corresponding author: Lan Du)

while the kernels at larger scales are more responsible for higher-level features. Therefore, the multi-scale CFA model will show more comprehensive information of SAR images compared with the CFA model with convolution kernels at a single scale. Moreover, we consider to combine the multi-scale CFA model and latent variable support vector machine (LVSVM) [12] into a united framework. The LVSVM is linked to the factors learned from multi-scale CFA model, yielding max-margin discrimination, as the recognition criterion in the feature space. The interplay between the likelihood function of multi-scale CFA model and maximum margin constraint induced by LVSVM can yield latent representations more discriminative and reasonable for supervised prediction tasks. Experimental results on MSTAR dataset show that the learned convolution kernels and factors can describe SAR information well and the proposed model has excellent classification performance.

The rest of the paper is structured as follows. We will have a brief review of the FA model and LVSVM in Section 2. Further, we introduce our MMCFA model in Section 3. Experimental results comparing with other methods of SAR ATR are shown in Section 4. Finally, we summarize this paper in Section 5.

2. BACKGROUND

2.1. FA model

Denote the N observed data set by $\boldsymbol{X}=\{\boldsymbol{x}_i\}_{i=1}^{N}$. The FA model assumes that $\boldsymbol{x}_i$ is generated via a linear mapping from a low-dimensional latent variable plus an observation noise, and $\boldsymbol{x}_i$ can be represented as follows

$$\boldsymbol{x}_i = \boldsymbol{D}\boldsymbol{z}_i + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_i \tag{1}$$

where $\boldsymbol{\mu}$ is the mean vector of the data; $\boldsymbol{z}_i$ is a latent factor and assumed Gaussian distributed as $\mathrm{N}(\boldsymbol{z}_i|0,\mathbf{I})$ with $\mathrm{N}(\bullet)$ denoting the Gaussian distribution; $\boldsymbol{D}$ is the factor loading matrix that maps the latent factor $\boldsymbol{z}_i$ to the observed data $\boldsymbol{x}_i$; the noise variable $\boldsymbol{\varepsilon}_i$ is assumed as $\mathrm{N}(\boldsymbol{\varepsilon}_i|0,\boldsymbol{\Psi})$. The noise covariance $\boldsymbol{\Psi}$ is a diagonal matrix and can be thought of as being sensor noise. Thus, $\boldsymbol{x}_i \sim \mathrm{N}(\boldsymbol{\mu},\boldsymbol{D}\boldsymbol{D}^{\mathrm{T}}+\boldsymbol{\Psi})$.

2.2. LVSVM

Given a labeled training dataset $\{(\boldsymbol{x}_i,y_i)|y_i\in\{-1,+1\}\}_{i=1}^{N}$, support vector machines (SVM) describes a binary linear recognition with the decision function $\hat{y}_i=\mathrm{sign}(\boldsymbol{\omega}^{\mathrm{T}}\tilde{\boldsymbol{x}}_i)$, where $\boldsymbol{\omega}$ is the weighted coefficient and $\tilde{\boldsymbol{x}}_i=[\boldsymbol{x}_i;1]$ is augmented feature vector. $\boldsymbol{x}_i$ is classified as a positive sample $(+1)$ when $\hat{y}_i\ge 0$, while as a negative sample (-1) in case of $\hat{y}_i<0$. SVM is a max-margin method, in which the margin is defined as the smallest distance between the decision boundary and any of the samples. To maximize the margin, SVM deals with the problem

$$\begin{aligned}\min_{\boldsymbol{\eta},\xi_n}\ & \frac{1}{2}\|\boldsymbol{\omega}\|_2^2 + C_0\sum_{i=1}^{N}\xi_i\\ s.t.\quad & y_i\boldsymbol{\omega}^{\mathrm{T}}\tilde{\boldsymbol{x}}_i \ge 1-\xi_i\\ & \xi_i\ge 0, i=1,\ldots,N\end{aligned} \tag{2}$$

where C_0 is a positive tuning parameter. Problem Eq. (2) can be solved by convex optimization algorithm.

Polson et al [12]. present a latent variable representation of SVM (LVSVM). They use the complete likelihood function of the weighted coefficient of SVM to represent the entire optimization function, so that SVM can be combined with a probabilistic model. Specifically, the optimization of Eq. (2) mode is equivalent to finding the following coefficient of pseudo-posterior distribution

$$p(\boldsymbol{\omega}|\boldsymbol{y}) \propto \exp(-d(\boldsymbol{\omega})) \propto C_\alpha \phi(\boldsymbol{y}|\boldsymbol{\omega})p(\boldsymbol{\omega}) \tag{3}$$

where $\boldsymbol{y}=[y_1,y_2,...,y_N]$; C_α is the normalized constant of pseudo-posterior distribution; $p(\boldsymbol{\omega})$ is prior distribution of coefficient $\boldsymbol{\omega}$; $\phi(\boldsymbol{y}|\boldsymbol{\omega})$ is the pseudo-likelihood contribution, and can be expressed as

$$\phi(\boldsymbol{y}|\boldsymbol{\omega})=\prod_i \phi(y_i|\boldsymbol{\omega}) = \exp\left(-2C_0\sum_{i=1}^{N}\max\left(1-y_i\boldsymbol{\omega}^{\mathrm{T}}\tilde{\boldsymbol{x}}_i,0\right)\right) \tag{4}$$

3. MULTI-SCALE CONVOLUTIONAL FACTOR ANALYSIS MODEL WITH MAX-MARGIN CONSTRAINT

Consider N SAR samples $\{\boldsymbol{x}_i\}_{i=1}^{N}$, with $\boldsymbol{x}_i\in\mathbb{R}^{N_x\times N_y}$, where N_x and N_y represent the number of pixels. We consider each $\boldsymbol{x}_i$ is expanded in terms of convolution kernels. Let $\boldsymbol{D}=\left\{\boldsymbol{d}_k^l\in\mathbb{R}^{N_x^l\times N_y^l}\right\}_{k=1,l=1}^{K,L}$ be the multi-scale convolution kernel bank with L different scales, and K kernels per scale, where $N_x^l \ll N_x$ and $N_y^l \ll N_y$. Define $\boldsymbol{Z}=\{\boldsymbol{Z}_i\}_{i=1}^{N}$ as the set of factors, where $\boldsymbol{Z}_i=\left\{\boldsymbol{Z}_{i,k}^l\in\mathbb{R}^{(N_x-N_x^l+1)\times(N_y-N_y^l+1)}\right\}_{k=1,l=1}^{K,L}$ consists of $K\times L$ factors maps for the representation of $\boldsymbol{x}_i$. So the $\boldsymbol{x}_i$ can be represented as:

$$\boldsymbol{x}_i = \sum_{l=1}^{L}\sum_{k=1}^{K}\boldsymbol{d}_k^l * \boldsymbol{Z}_{i,k}^l + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_i \tag{5}$$

where $*$ is the convolution operator; $\boldsymbol{\mu}$ is the mean matrix of the sample; $\boldsymbol{\varepsilon}_i\in\mathbb{R}^{N_x\times N_y}$ is the noise matrix. Since the size of all factor weight $\boldsymbol{Z}_i$, i.e.,

$(N_x - N_x^l + 1) \times (N_y - N_y^l + 1) \times \mathrm{K} \times \mathrm{L}$, is much larger than that of the input $\boldsymbol{x}_i$, i.e., $N_x \times N_y$, our model is overcomplete and runs the risk of having infinite solutions for the convolution kernels, where some of them can not reflect the inputs' structures. Therefore, in this model, we perform max-pooling on each factor weight $\boldsymbol{Z}_{i,k}^l$, where all elements of $\boldsymbol{Z}_{i,k}^l$ are set to zero other than the maximum one in the pooling region, to facilitate the learning of the convolution kernels with certain structures. A similar method was also adopted in [13].

Assume that a label $y_i \in \{1, \ldots, C\}$ is associate with each of the N SAR samples, so that the training set may be denoted $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$. We wish to learn a classifier that maps the factors $\boldsymbol{Z}_i = \{\boldsymbol{Z}_{i,k}^l\}_{k=1,l=1}^{K,L}$ to an associated label y_i. The $\boldsymbol{Z}_i$ can be unfolded into the vector $\boldsymbol{z}_i$. We desire the classifier mapping $\boldsymbol{z}_i \to y_i$ and our goal is to learn the multi-scale CFA model and classifier jointly.

As mentioned in Section 1, LVSVM is chosen as the classifier, since it is able to model label $\boldsymbol{y}$ in the hinge loss function under Bayesian framework and has the successful behavior of max-margin models in recognition. And the construction in Eq. (5) may be viewed as a special class of FA model. We rewrite Eq. (5) in Eq. (6). The MMCFA model structure can be expressed as

$$
\begin{aligned}
&\boldsymbol{x}_i = \sum_{l=1}^{L}\sum_{k=1}^{K}\sum_{s=1}^{S} z_{i,k,s}^l \boldsymbol{d}_{k,s}^l + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_i \\
&z_{i,k,s}^l \sim \mathcal{N}(0, 1/\alpha_{i,k,s}^l), \alpha_{i,k,s}^l \sim \mathrm{Ga}(a_0, b_0), \\
&d_{k,j}^l \sim \mathcal{N}(0, 1/\beta_{k,j}^l), \beta_{k,j}^l \sim \mathrm{Ga}(c_0, d_0), j = 1, \ldots J_l^2, \\
&\boldsymbol{\varepsilon}_i \sim \mathcal{N}(0, \gamma_i^{-1}\mathbf{I}), \gamma \sim \mathrm{Ga}(e_0, f_0), \\
&y_i, \lambda_i | \boldsymbol{\omega}, \tilde{\boldsymbol{w}}_i \sim \phi(y_i, \lambda_i | \boldsymbol{\omega}, \tilde{\boldsymbol{w}}_i), \\
&\boldsymbol{\omega} | \sigma \sim \mathcal{N}(0, \sigma^{-1}\mathbf{I}), \sigma \sim \mathrm{Ga}(\tau_1, \tau_2)
\end{aligned} \tag{6}
$$

where $\boldsymbol{d}_{k,s}^l \in \mathbb{R}^{N_x \times N_y}$ represents a shifted version of $\boldsymbol{d}_k^l$. In detail, $\boldsymbol{d}_k^l$ is firstly padded with zeros to scale $N_x \times N_y$, and then $\boldsymbol{d}_{k,s}^l$ is generated via circularly shifting the elements in the zero-padded matrix with all S possible shifts. J_l ($J_l = N_x^l \times N_y^l$) denotes the number of elements in the convolution kernels $\boldsymbol{d}_k^l$, and $d_{k,j}^l$ is the jth element of $\boldsymbol{d}_k^l$; a_0, b_0, c_0, d_0, e_0, and f_0 are preset hyperparameters. $\tilde{z}_i = [\boldsymbol{z}_i; 1]$ is the augmented feature vector of $\boldsymbol{w}_i$. In MMCFA, we consider the latent representation $\boldsymbol{w}_i$ as the input features to learn a discriminative latent space described by $\boldsymbol{D}$. In case of small sample size, $\boldsymbol{z}_i$ is expanded by $\boldsymbol{Z}_i^{'}$ ($\boldsymbol{Z}_i^{'} = [\sum_{k=1}^{K} \boldsymbol{Z}_{i,k}^1; \sum_{k=1}^{K} \boldsymbol{Z}_{i,k}^2; \ldots; \sum_{k=1}^{K} \boldsymbol{Z}_{i,k}^L]$) rather than $\boldsymbol{Z}_i$ in order to solve the fitting problem. All the unknown parameters in Eq. (6) can be estimated via a Markov chain Monte Carlo (MCMC) method based on Gibbs sampling.

4、EXPERIMENTS

4.1. Data description

In this subsection, MSTAR dataset with three-target data is tested using our proposed model. The data contains two types of tanks, T72 and BMP2, and an vehicle BTR70. Each of the target has full aspect coverage from 0° to 360° and different views at 15° and 17° depression angles. The data at depression 17° are used for training and those at depression 15° for test. The number of aspect views available for these targets is listed in Table 1.

4.2. Recognition results

In this subsection, firstly we compare the recognition performance of our proposed MMCFA model in different parameter setting on MSTAR dataset, including 1-MMCFA (our model with single scale) and 2-MMCFA (our model with two scales). All recognition results of each model are the highest accuracy in the optimal parameter setting. As shown in Table 2, by comparing the first row with the third row and the second row with the fourth row, accuracies are improved due to the increase of convolution kernels with different scales. As shown in Fig. 1, the multi-scale convolution kernels learned via 2-MMCFA (13×13, 23×23) are able to extract comprehensive information, thus offering the potential to improve the separability of the factors in recognition task.

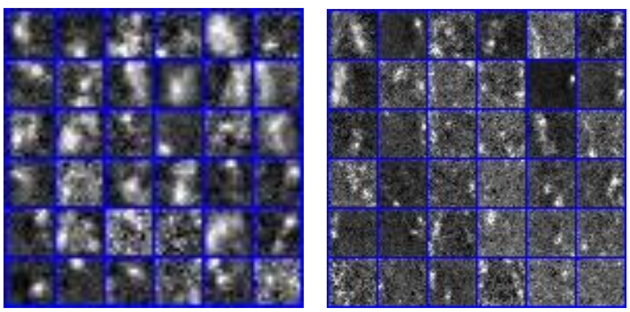
Secondly, we compare the proposed model with some other SAR ATR methods. The correct recognition rates of these methods are shown in Fig. 2. In Fig. 2, "2-MMCFA" represents our proposed model; "SVM" represents directly using original images as input features in linear SVM; "JSR" represents joint Sparse Representation [4]; "PCA" represents PCA with kernel SVM [3]; "HD" represents Particle Swarm Optimization with Hausdorff Distance [8]; "AE-CNN" represents CNN with sparse AE pretraining [11]; "Euclidean distance AE" represents AE combined with a supervised constraint by a Euclidean distance restriction on AE [1]; "sparse AE" presents AE model with sparse constraint and other preprocessing methods, such as filtering and image segmentation [14]. As shown in Fig.2, compared to some traditional methods (JSR, PCA and HD), our proposed MMCFA model has achieved the highest accuracy due to the exploitation of multi-scale convolution operator and supervised restriction. Compared to these deep learning methods (sparse AE, AE-CNN and Euclidean Distance AE), the accuracy of our proposed model is also better.

Table 1 Type and number of training and test samples for three-target dataset

Dataset	BMP2			BT R70	T72		
	C21	9566	9563	C71	132	S7	812
Training	233	0	0	233	232	0	0
Test	196	196	195	196	196	191	195

Table 2 Accuracies of our proposed model in different parameter set for MSTAR dataset

Vehicle Model	BMP2	BTR2	T72	Average
1-MMCFA (19×19)	92.45%± 0.38	94.27%± 1.61	88.32%± 1.53	90.95%±1.03
1-MMCFA (23×23)	91.65%± 0.57	92.86%± 0.92	90.55%± 1.33	91.21%±0.94
2-MMCFA (9×9, 19×19)	93.92%± 0.96	**95.66%**± 1.07	90.58%± 2.14	92.75%±1.41
2-MMCFA (13×13, 23×23)	**95.40%**± 1.57	94.90%± 0.39	**93.81%**± 0.43	**94.67%**±0.42



(a) (b)

Fig. 1. The learned convolution kernels for SAR images. (a) 36 small scale kernels of size 13×13. (b) 36 large scale kernels of size 23×23.

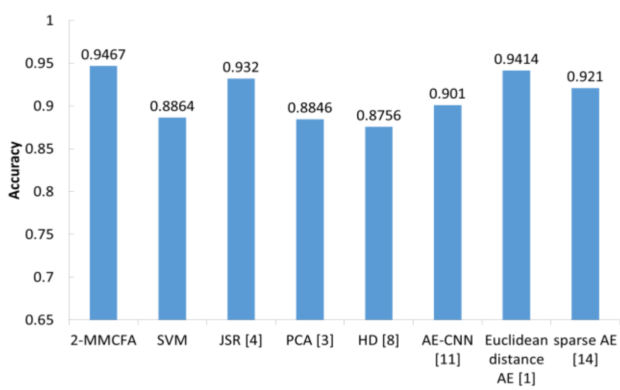


Fig. 2. 3-target accuracies obtained by different SAR ATR methods for MSTAR dataset

5. CONCLUSION

In this paper, we propose an improved FA model, namely, MMCFA model, for SAR automatic target recognition. The MMCFA model is a combination of the multi-scale CFA model and LVSVM. To validate the proposed model, it is compared with some traditional recognition methods on MSTAR dataset. Based on the experimental analysis on the recognition accuracy, the proposed model obtains excellent recognition performance.

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